

Question 1 : Define Data Quality in the context of ETL pipelines. Why is it more than just data cleaning?

Data quality in ETL pipelines refers to the accuracy, completeness, consistency, timeliness, and validity of data as it moves from source systems to a target warehouse, ensuring it is "fit for purpose" for analytics and decision-making.

Important because :-

- **Data Integration** - ETL collects data from different sources (like databases, files, apps) and combines it into one place so everything can be viewed together easily.
- **Data Quality** - ETL cleans and fixes raw data by removing errors and making it consistent, so the data becomes more accurate and reliable.
- **Important for Data Warehousing** - ETL prepares and organizes data properly before storing it in a data warehouse, so it is ready for analysis and reporting.
- **Better Decision-Making** - ETL helps businesses understand their data clearly, which supports better planning, forecasting, and smarter decisions.
- **Operational Efficiency** - ETL automates data processing, making it faster. This helps organizations access updated data quickly and make real-time decisions.

Question 2 : Explain why poor data quality leads to misleading dashboards and incorrect decisions.

Here is why poor data quality ruins dashboards and decisions:

- Inaccurate Data Skews Reality
- Incomplete Data Creates Blind Spots
- Inconsistent Data Destroys Trust
- Outdated Data Misguides Strategy
- Hidden Data (Outliers) Distort Metrics

Question 3 : What is duplicate data? Explain three causes in ETL pipelines.

Duplicate data means the same record appears more than once in a dataset or system.

Instead of having one correct record, the same customer, transaction, or ID is stored multiple times.

This causes wrong reports and inaccurate results.

In ETL (Extract, Transform, Load), duplicates are a common problem that can distort dashboards and analytics.

Common Causes of Duplicate Data in ETL

- **Merging Data from Multiple Sources**

When data is collected from different systems (like CRM and marketing tools), the same person may exist in both systems. If the ETL process does not recognize they are the same person, it stores them twice.

Example:

John Doe exists in both systems → After loading, he appears twice in the data warehouse.

- **Re-running ETL Jobs (Retry Problem)**

If an ETL job fails and is restarted, it may load the same data again. If the system is not designed properly, it will insert the same records twice.

Example:

10,000 records are loaded. Job fails and runs again → Same 10,000 records are added again → Now 20,000 records.

- **Wrong Join Logic During Transformation**

When combining tables, if the join key is not unique, it can create extra duplicate rows.

Example:

If one customer ID appears multiple times in the customer table, joining it with orders will duplicate order records.

Question 4 : Differentiate between exact, partial, and fuzzy duplicates.

Exact Duplicates - Records that are completely identical in all columns. Both rows are 100% the same.

ID	Name	Email
101	John Doe	john@email.com
101	John Doe	john@email.com

Partial Duplicates - Records that are not fully identical, but share some important fields (like same ID or email).

ID	Name	Email
101	John Doe	john@email.com
101	John D.	john@email.com

Fuzzy Duplicates - Records that are similar but not exactly the same, usually due to spelling mistakes or formatting differences.

Name	City
Jon Doe	New York
John Doe	NY

Question 5 : Why should data validation be performed during transformation rather than after loading?

- Prevents Garbage in the Database
- Saves Time and Effort
- Maintains Data Quality
- Supports ETL Efficiency

Question 6 : Explain how business rules help in validating data accuracy. Give an example.

Business rules are specific conditions or standards defined by a company to ensure data follows the organization's requirements.

They help in checking if data is correct, complete, and meaningful before it is used for reporting or analysis.

Example

Business Rule:

Sales_Amount > 0

Scenario:

- Record has Sales_Amount = -500
- Validation fails because negative sales are not allowed

- Correcting or rejecting this record ensures accurate revenue reporting.

Question 7 : Write an SQL query on Sales_Transaction to list all duplicate keys and their counts using the business key (Customer_ID + Product_ID + Txn_Date + Txn_Amount)

```
SELECT Customer_ID, Product_ID, Txn_Date, Txn_Amount, COUNT(*) AS
duplicate_count
```

```
FROM Sales_Transactions
```

```
GROUP BY Customer_ID, Product_ID, Txn_Date, Txn_Amount
```

```
HAVING COUNT(*) > 1;
```

Customer_ID	Product_ID	Txn_Date	Txn_Amount	duplicate_count
C101	P11	2025-12-01	4000	3

Question 8 : Enforcing Referential Integrity Assume the following Customer_Master table:

Identify Sales_Transactions.Customer_ID values that violate referential integrity when joined with Customers_Master and write a query to detect such violations.

```
SELECT DISTINCT st.Customer_ID FROM Sales_Transactions st
```

```
LEFT JOIN Customers_Master cm ON st.Customer_ID = cm.CustomerID
```

```
WHERE cm.CustomerID IS NULL;
```

Result :- C105 and C106 are not present in Customers_Master, so they violate referential integrity.